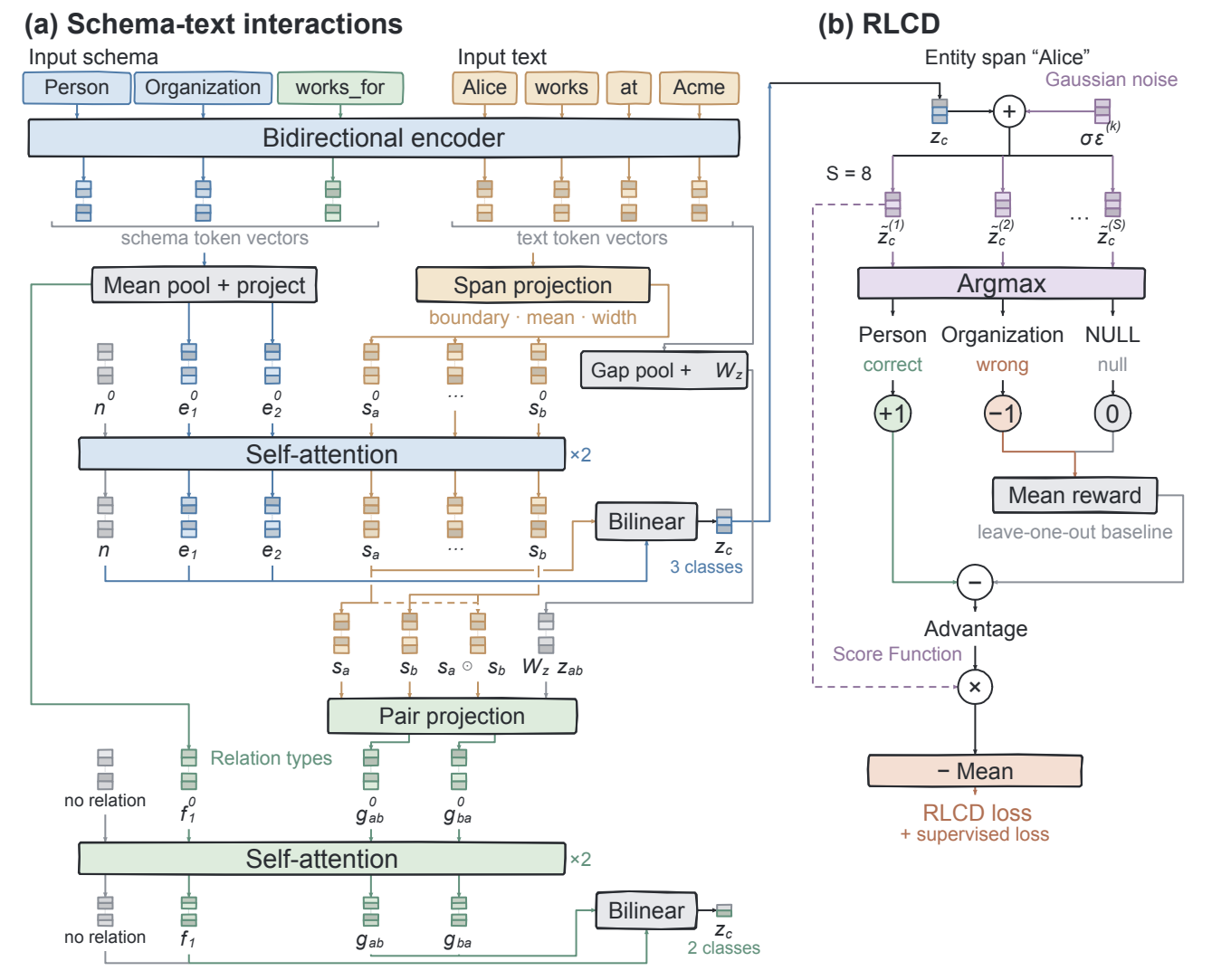

(a) Schema-text interactions
Input schema
Person
Organization
works_for
Input text
Alice
works
at
Acme
Bidirectional encoder
schema token vectors
text token vectors
Mean pool + project
Span projection
boundary · mean · width
Gap pool + W_z
Self-attention
×2
Bilinear
z_c
3 classes
Pair projection
Relation types
no relation
Self-attention
×2
Bilinear
z_c
2 classes
(b) RLCD
Entity span “Alice”
Gaussian noise
S = 8
Argmax
Person
Organization
NULL
correct
wrong
null
+1
−1
0
Mean reward
leave-one-out baseline
Advantage
Score Function
− Mean
RLCD loss
+ supervised loss